



TRUTHLENS AI  
VERIFY EVERY PIXEL



Executive Reports

Synthetic Politician Statement (Deepfake)



TRUTHLENS AI FORENSICS SUITE

Synthetic Politician Statement  
(Deepfake)

CASE\_IDENTIFIER: TL-8204

DIGITAL SIGNATURE HASH

# SHA-256: F0A79E2BTL-8204C94

COMPILED: 7/7/2026, 2:52:15 AM

EXECUTIVE RISK  
METRIC

92% CRITICAL  
THREAT

PIXEL AUTHENTICITY  
RATIO

8% UNALTERED  
ELEMENTS

ANALYSIS  
CONVERGENCE

96% SYSTEM  
STABILITY

I. EXECUTIVE SUMMARY

Comprehensive spectral and biometric analysis confirms that this image is a highly advanced synthetic manipulation. The face of the subject has been generated or swapped onto a standard briefing room backdrop using diffusion models. 92% risk assessment is driven by severe facial landmark light vector misalignment combined with telltale high-frequency synthetic noise structures.



II. CHRONOLOGICAL INCIDENT TIMELINE

PHASE\_1: DIGITIZATION\_EVENT

Metadata timestamp recorded in EXIF block  
2026-07-06 21:05:11 UTC (Stripped)

PHASE\_2: SECURE INGESTION

TruthLens Agent pins and seals hash registry  
7/7/2026, 2:52:15 AM

IBM WATSONX\_OK

v2.4.0

SECURE\_SHIELD: ACTIVE



III. IBM GRANITE REASONING SYNTHESIS

CORE COGNITIVE DEDUCTION:

Comprehensive spectral and biometric analysis confirms that this image is a highly advanced synthetic manipulation. The face of the subject has been generated or swapped onto a standard briefing room backdrop using diffusion models. 92% risk assessment is driven by severe facial landmark light vector misalignment combined with telltale high-frequency synthetic noise structures.

MATHEMATICAL FINDINGS REGISTER:

- Fourier spectral analysis peak at  $f_s = 0.34$  indicating GAN lattice structure.
- Facial light vectors deviate by 75 degrees from background scene illumination.
- Total absence of standard camera metadata (EXIF/JFIF header blocks).
- Concentric eye-reflection analysis shows physically impossible light refraction paths.

IV. VISUAL FINDINGS REGISTER

|                                                                                                                                                                                                                |                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Noise                                                                                                                                                                                                          | HIGH_SEVERITY   |
| <b>High-Frequency GAN Noise Pattern</b><br>Fourier transform analysis reveals periodic checkerboard artifacts characteristic of Generative Adversarial Networks (GAN) upscaling.                               |                 |
| Lighting                                                                                                                                                                                                       | HIGH_SEVERITY   |
| <b>Facial Landmark Lighting Discrepancy</b><br>Illumination angle on the nasal bridge (120°) is mathematically inconsistent with the ambient background light source (45°).                                    |                 |
| Artifacts                                                                                                                                                                                                      | MEDIUM_SEVERITY |
| <b>Pupil Asymmetry &amp; Irregular Refraction</b><br>Microscopic iris analysis indicates non-concentric pupil boundaries and symmetrical corneal light reflections, a common error in synthetic eye rendering. |                 |
| Compression                                                                                                                                                                                                    | HIGH_SEVERITY   |
| <b>Localized Compression Anomalies</b><br>Error Level Analysis (ELA) exhibits highly concentrated error layers specifically around the mouth and eyes, suggesting post-process overlay.                        |                 |

V. RETRIEVED RAG KNOWLEDGE INDEX

KB-101

RAG\_HIT: 99% Match

**Facial Synthetics & GAN Signature Analysis**  
Generative models produce systematic grid patterns in deep convolutional layers. These footprints show up as spectral peaks in the 2D Fourier power spectrum.  
SOURCE: IBM watsonx.ai Security Research Group (Ref: SD-2025-X)

KB-102

RAG\_HIT: 94% Match

**JPEG Quantization Discrepancy (Double Compression)**  
When a fake segment is pasted onto an existing JPEG, it goes through a double-compression cycle. The secondary quantization table differs from the base matrix, creating detectable ELA peaks.  
SOURCE: Granite Forensics RAG Index / IEEE Signal Processing Library

KB-103

RAG\_HIT: 91% Match

**Biometric Inconsistencies in Deep Generative Media**  
Synthetic media generators do not model detailed physiological optics correctly. Inconsistencies include lack of pulse-induced skin micro-color shifts and pupillary non-reactivity.  
SOURCE: DARPA MediFor Project Documentation Archive

VI. STRATEGIC REMEDIATION RECOMMENDATIONS

- 1. Flag file as highly synthetic/manipulated across media distribution protocols.
- 2. Issue warning: Facial audio alignment should be cross-examined with physical acoustic signatures.
- 3. Isolate face coordinates for deep facial-geometry reconstruction models.

COGNITIVE TRUST VERDICT SYSTEM

This document serves as an analytical cyber-dossier verifying image integrity. The observations generated are synthesized from automated multi-spectral filters combined with Granite-13B neural model outputs.

**IBM watsonx Core Autopilot**  
AUTHORIZED SECURITY SEAL



VERIFIED